



Ultrahigh-temperature ceramics: Empowering better energy usage in next-generation systems

By Kartik Nemani



Operating at higher temperatures can significantly enhance system efficiency in advanced aerospace, energy, and

thermal systems. But operating at these higher temperatures requires materials that can withstand increased atmospheric stresses, such as larger thermal gradients and accelerated oxidation mechanisms.

Ultrahigh-temperature ceramics (UHTCs) can enable components to withstand these stresses while maintaining functionality.¹ These ceramics have melting points above 3,000°C, and they offer exceptional oxidation resistance, thermal shock tolerance, and chemical stability. Such qualities make UHTCs ideal for thermal protection systems in hypersonic vehicles, spacecraft, nuclear fuel claddings, and turbine components.

Despite their potential, UHTCs face significant barriers to widespread adoption. For example, UHTC production typically requires extremely high sintering temperatures (about 2,000–2,200°C), which complicates controlled grain growth and porosity and consumes significant amounts of energy. Furthermore, many UHTCs have densities above 12 g/cm³, which is significantly higher than aerospace-grade alloys. This extra mass is untenable in weight-sensitive systems such as reentry vehicles or satellites, where every additional kilogram can translate to substantial increases in launch costs and fuel requirements.

While these drawbacks are nontrivial, recent innovations offer viable solutions that justify continued investment in UHTCs. For example, advancements

in additive manufacturing and electric-field-assisted sintering methods enable the fabrication of architected UHTCs with myriad structures, which help reduce material usage while preserving thermal performance and an optimized strength-to-weight ratio.

Ceramic-metal composites and fiber-reinforced UHTCs combine the best of both worlds: the toughness and lower density of metals with the thermal resilience of ceramics. Likewise, functionally graded materials tailor properties across a component by using UHTCs only where necessary, such as in the outer skins or hot zones of aircraft, while maintaining lighter support structures internally.

Emerging material classes such as high-entropy ceramics and MAX phases offer lower thermal conductivity, enhanced oxidation resistance, and improved processability compared to traditional UHTCs. Additionally, 2D ceramic materials can serve as UHTCs with added capabilities, such as electromagnetic interference shielding and energy harvesting.²

Thanks to these advancements, even if UHTC components remain heavier or more costly to produce, their benefits to overall system-level efficiency far outweigh the material penalties. In aerospace and nuclear power, where operational efficiency and reliability are mission critical, these paybacks can be transformative.

Ultimately, the future of sustainable engineering lies not just in new fuels or engines but in materials that perform under extreme conditions. Despite challenges, UHTCs are poised to be the thermal backbone of next-generation, energy-efficient systems. With innovative processing methods and structural designs, they shift from burdens to strategic assets enabling—not just withstanding—high-efficiency performance.

UHTC development at DOE National Labs

The perspective below is from Jorgen Rufner, senior materials scientist and Advanced Manufacturing Group Lead at Idaho National Laboratory

“Smart selection and strategic use of UHTC components are essential, especially as more technologies demand materials that can survive extreme environments. However, UHTCs often depend on rare or specialized elements and energy-intensive processing, making them vulnerable to supply chain disruptions driven by geopolitical and resource constraints.

These limitations highlight the need for resilient supply strategies and diversified sourcing, an area where U.S. Department of Energy National Labs are actively developing solutions in large-scale manufacturing and energy-efficient materials by design.”

References

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